SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario-Based

Code of Conduct:

I Muthu Krishnan K(192511039) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Muthu Krishnan K
Signature of the student:

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.

4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

QUESTION 1

Smart Campus Network Design Using IPv4 Subnetting

Scenario Understanding

A Smart Campus has been allocated the IP block **192.168.100.0/24**. Three departments require separate subnetworks:

- School of Computing → /25
- School of Electronics → /26
- School of Mechanical Engineering → /27

The objective is to divide the given network efficiently while ensuring sufficient host addresses for each department.

Step 1: School of Computing (/25)

Subnet Mask

/25 = 255.255.255.128

Network Address

192.168.100.0

Broadcast Address

192.168.100.127

Host Range

192.168.100.1 – 192.168.100.126

Number of Hosts

$2^{(32-25)} - 2$

$= 2^7 - 2$

$= 128 - 2$

= 126 usable hosts

Decision

This subnet is suitable for the School of Computing because it supports a large number of computers, servers, and networking devices.

Step 2: School of Electronics (/26)

Subnet Mask

255.255.255.192

Network Address

192.168.100.128

Broadcast Address

192.168.100.191

Host Range

192.168.100.129 – 192.168.100.190

Number of Hosts

2^(32−26) − 2

= 64 − 2

= 62 usable hosts

Decision

This subnet provides adequate space for laboratories and faculty systems.

Step 3: School of Mechanical Engineering (/27)

Subnet Mask

255.255.255.224

Network Address

192.168.100.192

Broadcast Address

192.168.100.223

Host Range

192.168.100.193 – 192.168.100.222

Number of Hosts

2^(32−27) − 2

= 32 − 2

= 30 usable hosts

Decision

This subnet is appropriate for a department with fewer devices.

Remaining Unused Address Space

Allocated Range:

192.168.100.0 – 192.168.100.223

Remaining Range:

192.168.100.224 – 192.168.100.255

Total Remaining Addresses:

32

Conclusion

The subnet allocation successfully divides the network into three independent departments while minimizing address wastage. This follows standard subnetting principles used in enterprise network design.

QUESTION 2

Multinational Software Company Network Design

Scenario Understanding

The company has received the private IP block:

10.10.0.0/16

Different departments require different subnet sizes.

Software Development Department (/24)

Network Address

10.10.0.0

Broadcast Address

10.10.0.255

Host Range

10.10.0.1 – 10.10.0.254

Usable Hosts

254

Justification

The Software Development team typically has the largest number of employees and devices.

Quality Assurance Department (/25)

Network Address

10.10.1.0

Broadcast Address

10.10.1.127

Host Range

10.10.1.1 – 10.10.1.126

Usable Hosts

126

Justification

Provides sufficient addresses for testing systems and staff workstations.

Human Resources Department (/26)

Network Address

10.10.1.128

Broadcast Address

10.10.1.191

Host Range

10.10.1.129 – 10.10.1.190

Usable Hosts

62

Justification

HR typically requires fewer devices than technical departments.

Executive Management Department (/27)

Network Address

10.10.1.192

Broadcast Address

10.10.1.223

Host Range

10.10.1.193 – 10.10.1.222

Usable Hosts

30

Justification

Executives require the smallest subnet because of the limited number of users.

Remaining Unused Address Space

Unused Range:

10.10.1.224 – 10.10.255.255

Conclusion

The network administrator efficiently utilized VLSM concepts to allocate address space according to organizational requirements.

QUESTION 3

IPv6 Migration in a Smart Hospital

Scenario Understanding

The hospital is transitioning from IPv4 to IPv6 to support thousands of medical devices and IoT systems.

Given IPv6 Address:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Compressed Form

Using IPv6 compression rules:

2001:db8:abcd::/64

2. Expanded Form

2001:0db8:abcd:0000:0000:0000:0000:0000/64

3. Network Prefix

Prefix Length:

/64

Network Portion:

2001:0db8:abcd:0000

Host Portion:

0000:0000:0000:0000

4. Number of Host Bits

$128 - 64$

$= 64$ bits

5. Total Host Addresses

2^{64}

$= 18,446,744,073,709,551,616$

addresses

Decision and Analysis

IPv6 provides an extremely large address space compared with IPv4, making it ideal for smart hospitals containing sensors, medical equipment, mobile devices, and future expansions. Large-scale IPv6 deployments commonly use /64 subnets because they provide ample addressing capacity and support standard IPv6 operations.

QUESTION 4

Routing Table Analysis

Scenario Understanding

The ISP router contains the following routing table:

- 192.168.1.0/24
- 192.168.2.0/24
- 192.168.3.0/24

Destination Address:

192.168.2.45

Step 1: Determine Destination Network

Network:

192.168.2.0/24

Since 192.168.2.45 lies between:

192.168.2.1 – 192.168.2.254

the packet belongs to:

192.168.2.0/24

Step 2: Matching Routing Table Entry

Matched Entry:

192.168.2.0/24

Step 3: Next-Hop Decision

Router forwards the packet through the interface connected to:

192.168.2.0/24

Reasoning

Routers compare destination addresses with routing table entries and select the best matching route.

Step 4: Default Route

If no route matches:

0.0.0.0/0

will be used.

Conclusion

The router successfully identifies the correct subnet and forwards the packet through the appropriate interface. Default routes are essential for handling unknown destinations.

QUESTION 5

Administration and Finance LAN Configuration

Scenario Understanding

A Layer-3 Router connects two LANs:

Administration LAN → 192.168.10.0/26

Finance LAN → 192.168.10.64/26

Administration LAN

Network Address

192.168.10.0

Broadcast Address

192.168.10.63

Host Range

192.168.10.1 – 192.168.10.62

Default Gateway

192.168.10.1

Suggested PC Addresses

PC-A1 = 192.168.10.2

PC-A2 = 192.168.10.3

PC-A3 = 192.168.10.4

Justification

All PCs belong to the same subnet and use the router interface as their gateway.

Finance LAN

Network Address

192.168.10.64

Broadcast Address

192.168.10.127

Host Range

192.168.10.65 – 192.168.10.126

Default Gateway

192.168.10.65

Suggested PC Addresses

PC-F1 = 192.168.10.66

PC-F2 = 192.168.10.67

PC-F3 = 192.168.10.68

Justification

All Finance hosts use the Finance router interface to communicate with external networks.

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